

0 4 . 1

The letters **P**, **Q**, **R**, **S** and **T** represent ways substances can move across membranes.

- **P** – diffusion through the phospholipid bilayer
- **Q** – facilitated diffusion
- **R** – active transport
- **S** – co-transport
- **T** – osmosis

For each of the following examples of transport across membranes, select the letter that represents the way in which the substance moves across the membrane.

Write the appropriate letter in each box provided.

[3 marks]

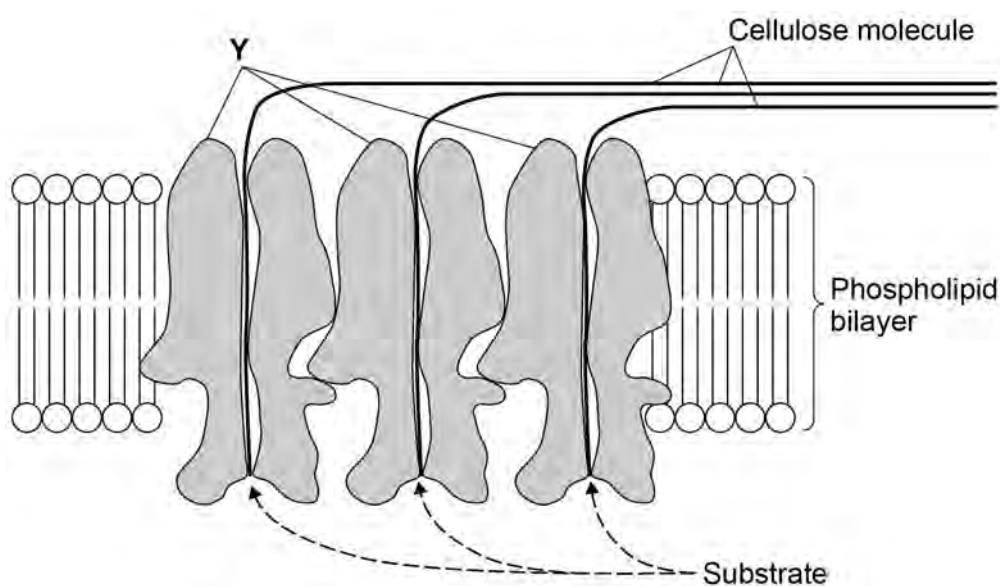
Transport through a channel protein

Transport of small, non-polar molecules

Transport of glucose with sodium ions

Figure 3 shows how a plant cell produces its cell wall.

Figure 3



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Y is a protein. One function of **Y** is to transport cellulose molecules across the phospholipid bilayer.

Using information from **Figure 3**, describe the other function of **Y**.

[2 marks]

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What is the evidence in **Figure 3** that the phospholipid bilayer shown is part of the cell-surface membrane?

[1 mark]

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In the cell wall, bonds hold the cellulose molecules together side by side.

Tick (✓) **one** box that describes the type of bond that holds the cellulose molecules together side by side.

[1 mark]

Ester

☐

Hydrogen

☐

Ionic

☐

Peptide

☐